User's Manual

comhand.c

date get
- Purpose: displays the current system date
- Example: Date is February 5th, 2026; User wants to get date
    - date is 02/05/26
    - inputs "date get"
        - outputs the date as 02 05 26

date set <month> <day> <year>
- Purpose: sets the system date <mm> <dd> <yy>
- Example: Date is February 5th, 2026; User wants to set date to September 13th, 2023
    - inputs "date get"
        - outputs the date as 02 05 26
    - inputs "date set 09 13 23"
        - system has date set to 09/13/23
    - inputs "date get"
        - outputs the date as 09 13 23

help
- Purpose: shows this list of available commands
    - date get
    - date set
    - help
    - shutdown
    - time get
    - time set
    - version
- Example: New user fixes the incorrect date, then shuts down the system once done
    - inputs "help"
        - outputs the list shown above
    - inputs "date get"
        - outputs the date as 09 13 23
    - inputs "date set 02 05 26"
        - system has date set to 02/05/26
    - inputs "date set"
        - outputs the date as 02 05 26
    - inputs "shutdown"
        - outputs a confirmation "y or n" prompt
        - inputs "y"
            - outputs a message confirming the shutdown

shutdown

- Purpose: shuts down the system and prompts for the confirmation
- Example: User wants to shutdown the system
    - inputs “shutdown”
        - outputs a confirmation “y or n” prompt
    - inputs “y”
        - outputs a message confirming the shutdown

time get

- Purpose: displays the current system time
- Example: Time is 5:20:35 pm (in UTC time); User wants to get time
    - inputs “time get”
        - outputs the time as 17 20 35

time set <hours> <minutes> <seconds>

- Purpose: sets the system time <hh> <mm> <ss> [in UTC time]
- Example: Time is 5:20:35 pm; User wants to set time to 7:30:00 pm
    - inputs “time get”
        - outputs the time as 17 20 35
    - inputs “time set 19 30 00”
        - System has time set to 7:30:00 pm
    - inputs “time get”
        - outputs the time as 19 30 00

version

- Purpose: shows build/version information
- Example: User wants to get the build and version information
    - inputs “version”
        - outputs the build, progress, and version of the project

Programmer’s Manual

krtc.h

int bcd_to_int(unsigned char bcd);

- Purpose: turns a binary coded decimal into an integer
- Params: *bcd* is an unsigned char representing a binary coded decimal
- Returns: an integer representing an unsigned char representing a binary coded decimal

int get_seconds(void);

- Purpose: gets time from the RTC and returns the current number of seconds
- Params: none
- Returns: an integer representing the current numbers of seconds in the RTC

int get_minutes(void);
- Purpose: gets time from the RTC and returns the current number of minutes
- Params: none
- Returns: an integer representing the current numbers of minutes in the RTC

int get_hours(void);
- Purpose: gets time from the RTC and returns the current number of hours
- Params: none
- Returns: an integer representing the current numbers of hours in the RTC

int get_day(void);
- Purpose: gets time from the RTC and returns the current number of days
- Params: none
- Returns: an integer representing the current numbers of days in the RTC

int get_month(void);
- Purpose: gets time from the RTC and returns the current number of months
- Params: none
- Returns: an integer representing the current numbers of months in the RTC

int get_year(void);
- Purpose: gets time from the RTC and returns the current number of years
- Params: none
- Returns: an integer representing the current numbers of years in the RTC

int int_to_bcd(int value);
- Purpose: turns an integer into a binary coded decimal
- Params: *value* is an integer
- Returns: an unsigned char representing an integer turned into a binary coded decimal

void set_seconds(int seconds);
- Purpose: sets the integer value representing the current seconds in the RTC time
- Params: *seconds* is an integer to be put into the RTC time seconds value
- Returns: none

void set_minutes(int minutes);
- Purpose: sets the integer value representing the current minutes in the RTC time
- Params: *minutes* is an integer to be put into the RTC time minutes value
- Returns: none

void set_hours(int hours);
- Purpose: sets the integer value representing the current hours in the RTC time
- Params: *hours* is an integer to be put into the RTC time hours value
- Returns: none

void set_day(int day);

- Purpose: sets the integer value representing the current days in the RTC time
- Params: *day* is an integer to be put into the RTC time day value
- Returns: none

void set_month(int month);

- Purpose: sets the integer value representing the current months in the RTC time
- Params: *month* is an integer to be put into the RTC time month value
- Returns: none

void set_year(int year);

- Purpose: sets the integer value representing the current years in the RTC time
- Params: *year* is an integer to be put into the RTC time year value
- Returns: none

urtc.h

define TIME_BUFFER_SIZE 11

- Purpose: array of size 11 used for get_time return message

define DATE_BUFFER_SIZE 11

- Purpose: array of size 11 used for get_date return message

enum Month

- Purpose: sets months as integers 1-12 for use in set_date method

void get_time(void);

- Purpose: gets the current RTC time and prints it to the comhand's COM1
- Params: none
- Returns: none

int set_time(int hours, int minutes, int seconds);

- Purpose: sets the RTC time for the hours, minutes, and seconds values (for UTC time)
- Params: *hours*, *minutes*, *seconds* are integers that replace the previous relevant values in RTC time
    - *hours* has range 0-23, *minutes* and *seconds* have range 0-59
- Returns: int 0 means the program succeeded; int -1 means an error occurred

void get_date(void);

- Purpose: gets the current RTC date and prints it to the comhand's COM1
- Params: none
- Returns: none

int set_date(int month, int day, int year);

- Purpose: sets the RTC date for the year, month, and day values
- Params: *year*, *month*, *day* are integers that replace the previous relevant values in RTC date
    - *year* has range 0-99, *month* has range 1-12 and *day* has range 1-31
        - *day* param range can vary based off *month* and *year* param range
- Returns: integer 0 if the program succeeded; integer -1 if an error occurred

<u>serial.c</u>

int serial_poll(device dev, char *buffer, size_t len)

- Purpose: takes keyboard input using polling technique, stores it to memory, and echos it back to the terminal.
    - Has unique cases for special characters such as Backspace, Escape, etc.
- Params: *dev* is a device figuring out which port to poll from for keyboard input, **buffer* is a char pointer that writes typed characters and stores them to an array (**buffer* is always null-terminated), *len* is used for the buffer size, setting a maximum boundary for the program to prevent overflow
- Returns: positive integer the number of read characters from the buffer, activated through either buffer being full or if the Enter key is pressed; integer 0 if no characters were read, integer -1 if error occurred

<u>comhand.h</u>

define WHITESPACE_CHARS " \r\n\t"

- defines what characters need to be split in the string tokenizer

define BUFFER_SIZE 100

- size of the buffer

<u>comhand.c</u>

void mpx_write(const char *s)

- Purpose: helper function that writes to the console
- Params: constant char pointer **s* functions as a string to print to the console
- Returns: none

void cmd_help(void)

- Purpose: lists user help commands
- Params: none
- Returns: none

void cmd_version(void)

- Purpose: shows our current MPX module, compiled build, date and time
- Params: none

- Returns: none

void comhand(void)
- Purpose: handles user commands
- Params: none
- Returns: none

Contributions

Error 404	*R1*
Evan	krtc.h, urtc.h, comhand
Ryan	serial_poll
Vicenzo	comhand, version_command, help_command
Brandon	user’s manual, programmer’s manual, contributions